

Tarsier: Recipes for Training and Evaluating Large Video Description Models

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Abstract

Generating fine-grained video descriptions is a fundamental challenge in video understanding. In this work, we introduce Tarsier, a family of large-scale video-language models designed to generate high-quality video descriptions. Tarsier employs CLIP-ViT to encode frames separately and then uses an LLM to model temporal relationships. Despite its simple architecture, we demonstrate that with a meticulously designed two-stage training procedure, the Tarsier models exhibit substantially stronger video description capabilities than any existing open-source model, showing a +51.4% advantage in human side-by-side evaluation over the strongest model. Additionally, they are comparable to state-of-the-art proprietary models, with a +12.3% advantage against GPT-4V and a −6.7% disadvantage against Gemini 1.5 Pro. When upgraded to Tarsier2 by building upon SigLIP and Qwen2-7B, it further improves significantly with a +4.8% advantage against GPT-4o. Besides video description, Tarsier proves to be a versatile generalist model, achieving new state-of-the-art results across nine public benchmarks, including multi-choice VQA, open-ended VQA, and zero-shot video captioning. Our second contribution is the introduction of a new benchmark – DREAM-1K (<https://tarsier-vlm.github.io/>) for evaluating video description models, consisting of a new challenging dataset featuring videos from diverse sources and varying complexity, along with an automatic method specifically designed to assess the quality of fine-grained video descriptions. We make our models and evaluation benchmark publicly available at <https://github.com/bytedance/tarsier>.

1 Introduction

Large-scale Video-Language Models (LVLMs) have made significant progress recently. These models combine visual encoders with large language models (LLMs) to achieve zero-shot video understanding capabilities. Proprietary models, such as GPT-4V [51] and Gemini [64, 58], are currently considered state-of-the-art. Open-source models [35, 4, 32, 26, 82] have achieved high scores on public video understanding benchmarks, especially in video question answering [78, 87, 77, 32, 45]. However, they have not yet demonstrated performance comparable to proprietary models in challenging open-ended generative tasks.

In this paper, we focus on *fine-grained video description*, a task that encapsulates some core challenges in video understanding. In this task, the model generates a long answer in free-form text

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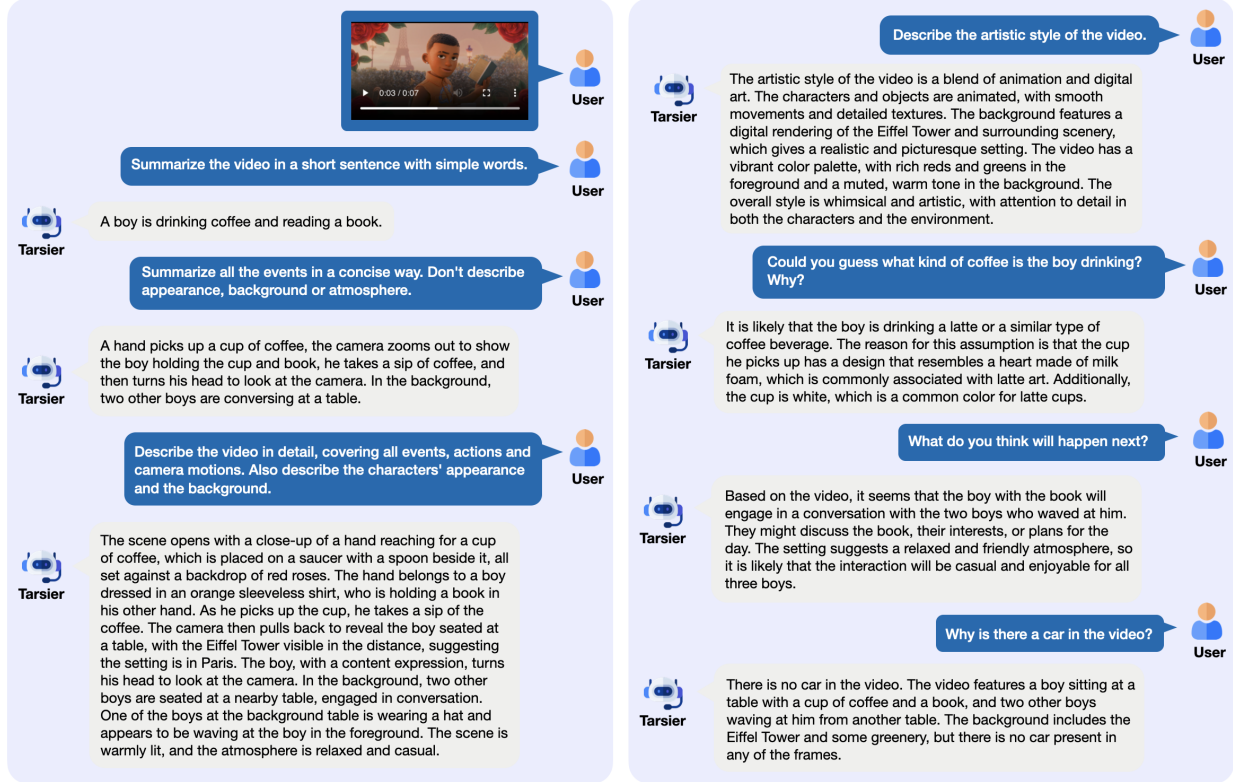


Figure 1: Example dialogue between a user and Tarsier, demonstrating the model’s ability to generate video descriptions with varying levels of detail. The model is also able to answer video-related questions in a zero-shot fashion. (Source movie: *Turning Red*, 2022)

to give a detailed coverage of the video content. The model must be both comprehensive and faithful, meaning that it must cover all notable events in the video without introducing hallucinations. The challenge increases when the video includes subtle movements, such as small but meaningful body actions, or quick motions. The challenge also increases as the video grows in complexity, capturing multiple events, subjects, and shots. Additionally, since the task does not have a unique answer, how to evaluate the model is a non-trivial question.

Empirically, we found that existing open source models lag far behind proprietary model in fine-grained video description. **Our main contribution** is the introduction of Tarsier models, a family of LVLMs with simple architectures that generate high-quality video descriptions (see Figure 1 for some examples). We trained two model sizes: 7B and 34B. Each Tarsier model consists of a CLIP-ViT encoder [56], a projection layer and a LLM. We train the model in two stages. The first stage is multi-task video-to-text pre-training, which enables the model to understand videos from different perspectives. The second stage is instruction tuning focused on generating multi-grained video descriptions. Both stages utilize high-dynamic videos with accurately matched text. Despite the simple architecture, we demonstrate that superior video description capability can be achieved by increasing the scale, diversity, and quality of the data.

Our second contribution is the introduction of a new benchmark for automatic evaluation of video description models. Existing video captioning datasets [9, 81, 72] typically pair each video

with a single-sentence caption. These captions are too short to stress-test the full capacity of models. We introduce a new video description dataset called DREAM-1K, which features videos of diverse sources and complexity. **DREAM-1K contains 1,000 video clips selected from five categories: live-action movies, animated movies, stock videos, YouTube videos, and TikTok-style short videos.** Typical DREAM-1K videos contain multiple events, possibly performed by multiple subjects in multiple shots, all carefully described in detail by human annotators. Additionally, we propose AutoDQ, an automatic evaluation method specifically designed to assess the quality of fine-grained video descriptions. AutoDQ breaks down the evaluation process into two steps: event extraction and entailment. This approach allows us to compute and compare the precision and recall of different models in an interpretable way.

Based on DREAM-1K, we demonstrate that Tarsier has video description capability comparable to state-of-the-art proprietary models and significantly outperforms any existing open-source model. Specifically, the Tarsier-34B model outperforms all open-source models as well as GPT-4V and Gemini 1.5 Pro in automatic evaluations. In human side-by-side evaluations, Tarsier-34B significantly outperforms the strongest open-source model with a +52% advantage, is favored over GPT-4V with a +12.3% advantage, and slightly falls behind Gemini 1.5 Pro with a −6.7% disadvantage. Moreover, Tarsier proves to be a versatile generalist model, achieving new state-of-the-art results on public benchmarks in multi-choice video Q&A [32, 77, 45], open-ended video Q&A [78, 24, 87], and zero-shot video captioning [9, 81, 72]. Finally, we present extensive ablation studies to gain insights into what makes a good LVLM. Our findings indicate that several factors contribute to the model’s strong performance: conducting large-scale multi-task pre-training, scaling up the LLM, tuning all LLM parameters, and fine-tuning the model with carefully annotated multi-grained video descriptions.

2 Related Work

Video-Language Modeling Video-language models typically consist of a visual encoder and a text decoder. On the encoder side, existing work either encodes video frames separately using a ViT [69, 12, 13], or encodes them jointly with architectures that can model spatial-temporal relationships [11, 79, 22, 54, 73]. For example, VALOR [11] employs a Video Swin Transformer [42], and Mirasol3B [73] uses TubeViT [53] to model the spatial-temporal relationship, while InternVideo2 [73] concatenates all the visual tokens and feeds them to a standard ViT. On the decoder side, standard Transformer [66] is usually used for autoregressive generation. Typical methods for pre-training the video-language model include video-text contrastive learning [80], masked video modeling [65], generative next-token prediction [2, 62, 69], or unifying multiple training objectives [70, 74, 73, 92]. Tarsier takes the simpler approach: it uses a frozen CLIP-ViT [56] to encode individual frames, then rely solely on the LLM for temporal modeling. The model is trained with the next-token prediction loss only.

Video-LLMs Recently, there has been a surge of research on how to connect a pre-trained video encoder to a pre-trained LLM to achieve zero-shot video understanding capabilities [31, 32, 44, 43, 90, 35, 4, 68, 82]. The two components are typically connected by either an adapter, such as Q-Former in BLIP2 [30], or a simple projection layer, such as a linear layer or MLP in LLaVA [38]. Tarsier adopts the later architecture. The combined model is usually fine-tuned on a relatively small instruction tuning set. For example, Video-LLaVA [35] is fine-tuned on 100K video-text pairs, while

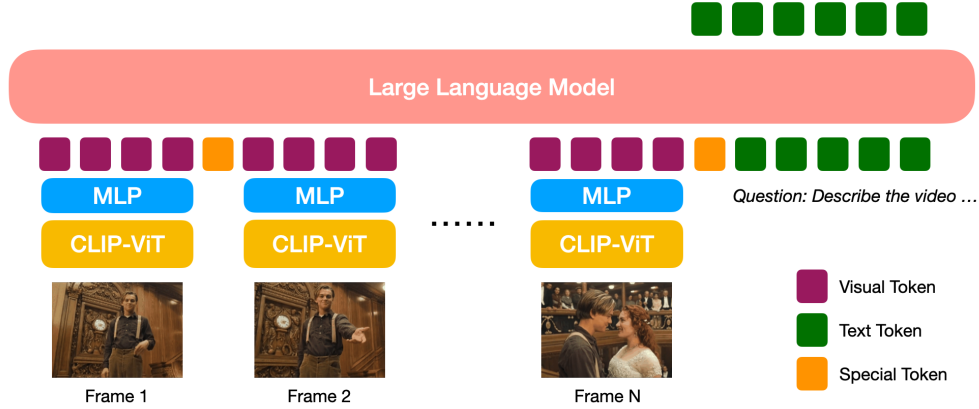


Figure 2: Tarsier model architecture. We use a frozen CLIP-ViT to encode frames, then project them to the LLM token space through an MLP. The LLM is then trained with the next-token prediction loss. (Source movie: *Titanic*, 1997)

VideoChat2 [32] and PLLaVA [82] are fine-tuned on 788K video-text pairs, including 7K detailed video descriptions. Another line of research explores training-free approaches that utilize existing image understanding models or short clip captioners to achieve video understanding [89, 26, 71, 25]. In contrast, our models are pre-trained with a larger and more diverse dataset, and then fine-tuned with high-quality, fine-grained description data.

Video Description Earlier research [69, 79, 22, 11, 13] typically pre-trains a video-language model and then fine-tunes it on specific video caption datasets, with performance measured by BLEU, ROUGE [36], METEOR [7], or CIDEr [67]. Although these fine-tuned models can achieve high scores on specific datasets, they often do not generalize well to open-world videos. Some recent models report zero-shot results [2, 84, 90, 92, 73]. However, since the standard video captioning benchmarks [9, 81, 72] are relatively simple, they do not stress-test the models’ full capacity. As video complexity grows and descriptions contain more details, traditional metrics cannot accurately reflect the discrepancy between the generated descriptions and the ground truth. Maaz et al.[44] propose using ChatGPT[1] for automatic evaluation, which has been widely used to evaluate open-ended question answering [35, 26, 68, 82]. We propose DREAM-1K and AutoDQ as a more challenging video description benchmark and a more interpretable method for automatic evaluation.

3 The Tarsier model

3.1 Model Architecture

Tarsier employs a simple model architecture. As depicted in Figure 2, it encodes each video frame separately. Initially, each frame is encoded using CLIP-ViT and then mapped to the downstream LLM’s token embedding space via a multi-layer perceptron (MLP). The resulting visual tokens are concatenated, separated by a special [end-of-image] token, allowing Tarsier to uniformly encode both videos and images. Then, all tokens are fed into the LLM for auto-regressive generation.

Dataset	Size	Task	Dataset	Size	Task
WebVid-10M [6]	2.8M	Video Captioning	LSMDC [59]	108K	Video Captioning
TGIF [33]	104K	Video Captioning	ActivityNet [27]	37K	Video Captioning
Charades [61]	16K	Video Captioning	Charades-Ego [60]	6K	Video Captioning
YouCook2 [93]	9K	Video Captioning	TACoS [57]	18K	Video Captioning
Ego4D [20]	1.08M	Video Captioning	Spoken Moments [49]	492K	Video Captioning
TREC-VTT [5]	64K	Video Captioning			
SSV2 [19]	168K	Action Recognition	HMDB [28]	6K	Action Recognition
Kinetics-700 [8]	536K	Action Recognition	COIN [63]	10K	Action Recognition
20BN-jester [46]	46K	Action Recognition	Multi-Moments [50]	1M	Action Recognition
FineAction [41]	82K	Action Recognition	RareAct [47]	2K	Action Recognition
DiDeMo [3]	41K	Action Localization	AVA [21]	27K	Action Localization
CLEVRER [86]	83K	Video QA	TGIF-QA [24]	72K	Video QA
EgoQA [18]	5K	Video QA			
Oops! [17]	14K	Intent Recognition	VideoInstruct [44]	100K	Video Instructions
VIST [23]	38K	Visual Storytelling			
ShareGPT4V [10]	94K	Image Captioning	MS COCO [37]	566K	Image Captioning
Flicker [55]	145K	Image Captioning	LLaVA-1.5 [38]	665K	Image Instructions
OpenOrca [34]	1M	Text	ShareGPT [16]	88K	Text

Table 1: Public datasets used in Tarsier pre-training.

This architecture captures the inter-frame relationship solely through the LLM, leveraging on its ability to do complex reasoning, which we believe is essential for understanding the connection between frames. During training, the CLIP-ViT component and the MLP are frozen, and the LLM is trained.

3.2 Training Recipe

We initialize Tarsier with the weights from the LLaVA-NeXT image models [39] and employ a two-stage training strategy¹. Initially, we conduct multi-task pre-training using large-scale, high-quality data. Subsequently, we perform instruction tuning on moderate-scale, human annotated data. We describe the training recipe as follows; for a detailed list of all the training hyperparameters, please see Appendix A.

3.2.1 Multi-task pre-training

In the pre-training stage, we trained our model across diverse tasks such as video captioning, video question answering, action recognition, multi-image understanding, and text generation, utilizing a broad array of public datasets as detailed in Table 1. To unify the training process, all tasks were converted into text generation tasks. For each dataset, we crafted specific prompts to define the output format. For instance, video captioning tasks may use prompts like *Describe the video in one sentence* or *Describe the video in detail*, tailored to the length of the annotated description; Video QA tasks may use prompts like *Give a short answer to the following question*. Once prompts were assigned to each dataset, we leveraged ChatGPT [1] to enhance prompt diversity. The model was

¹When building Tarsier2-7B, we employ SigLIP [88] as vision encoder and Qwen2 [85] as LLM. We conduct stage-0 (image-text alignment training) before the two-stage training strategy, and scale up the training data to over 38M, see <https://github.com/bytedance/tarsier> for more details.

subsequently trained to produce the annotated response based on the video/image input and the associated prompt. For datasets that have train/val/test splits, we use the training splits only.

Note that larger-scale video-text datasets such as HowTo100M [48], HD-VILA [83], and Panda-70M [14] exist. We opted not to use the video-text pairs in these datasets due to issues such as narrow domains or low-quality texts. We found that human-annotated descriptions of high-dynamic videos are the most helpful for training our models. **Notably, the WebVid-10M dataset [6] includes many instances where the video content can be described by just looking at a single frame, which are less valuable for video understanding.** To address this, we used instance re-sampling to prioritize videos involving human actions and animal movements, as these are more likely to feature dynamic events. This resulted in 2.8M video-text pairs from WebVid-10M being used in pre-training.

In addition to public datasets, we also added 3.5M in-house data to pre-training, including 2.4M high-quality video caption data similar to WebVid and 1.1M videos with object-tracking annotations. The later annotations are automatically generated for a subset of videos in WebVid-10M [6] and HD-VILA [83]. We use DEVA [15] to generate object-tracking bounding boxes for these videos at high frame rates, then train the model to predict the bounding box coordinates of every object in every frame.

Overall, our pre-training dataset comprises 13.6M video-text pairs. We trained the model on 48 A100 GPUs for 100K steps using a batch size of 288. Throughout the training process, **we randomly sampled between 8 to 32 frames per video.** To sample n frames, we first partitioned the video evenly into n segments, then randomly sampled one frame from each segment. Each sampled frame was resized to square before being sent to vision encoder.

3.2.2 Instruction Tuning

In the instruction tuning stage, we use 500K of in-house instruction tuning data. The dataset contains a wide spectrum of video complexities and diverse prompts. In addition to the tasks from pre-training, we added the following new tasks:

- **Describing Complex Videos at Different Levels of Granularity:** We collected a set of 100K movie clips featuring multiple shots, subjects, or events, and had annotators provide descriptions varying in length and detail, from brief motion summaries to comprehensive narratives of visual details. During training, we used different prompts to precisely specify the requirements for each level of granularity.
- **Describing Camera Motions:** We found that precisely describing camera motion is a weak point of existing models. To address this, we created a dataset rich in camera motions, including zooming, translating, panning, and rotating. Then we ask annotators to accurately describe these motions.
- **Creative Writing:** The model was also trained to generate creative outputs based on videos, such as imagining characters’ dialogues, drafting motivational speeches, and writing poems. We used self-instruct [75] to create diverse prompts and relied on GPT-4V [51] to generate responses based on human-annotated video descriptions.

By incorporating these tasks, we further enhanced the model’s capabilities in multi-grained video description. We fine-tuned the model for 10K steps on the instruction tuning set using 16 A100 GPUs and a batch size of 96. During training, we sampled 16 frames from each of the 100K complex

video clips mentioned above, while sampling 8 frames from other videos. This differentiation is necessary because movies often contain rapid actions and movements that can only be fully captured with a higher frame rate. The first and last frames of the video were always selected, with the intermediary frames evenly sampled throughout the video.

4 Evaluation of Video Description Models

In this section, we present DREAM-1K, a new benchmark and AutoDQ, an automatic evaluation method to comprehensive assess the quality of video description models.

4.1 DREAM-1K: A challenging video description benchmark

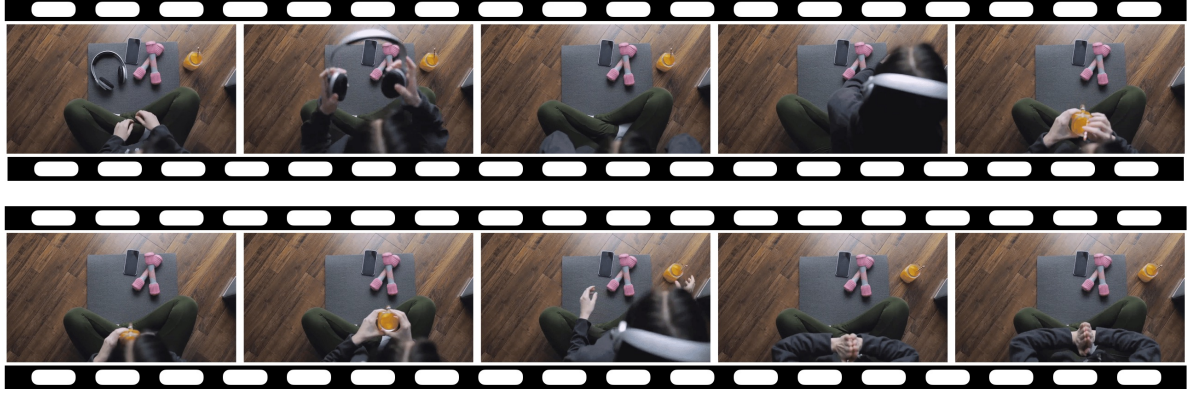
Generating high-quality video descriptions requires an understanding of multiple frames together. This understanding demands two primary capabilities. Firstly, the model should be capable of comparing frames at the pixel level to detect subtle motions, such as a person speaking or the camera panning. Secondly, it must link frames on a semantic level to reason about higher-level events, like the characters’ intentions and relationships. Current video captioning benchmarks, such as MSR-VTT [81] and VATEX [72], are relatively simplistic. Typically, each video in these benchmarks involves a single major action performed by one subject in a single shot. Additionally, the *single frame bias* [29] indicates that many actions in these datasets can be identified from still images, without the need for considering temporal relations.

	Live-action	Animation	YouTube	Shorts	Stock	Total
Number of videos	200	200	200	200	200	1,000
Avg. video length (second)	7.3	6.1	7.8	9.5	13.7	8.9
Avg. text length (word)	47.5	56.7	81.0	67.2	43.9	59.3
Avg. number of events	5.5	6.5	6.6	7.5	5.4	6.3
Avg. number of subjects	2.4	2.6	2.2	1.9	1.8	2.2
Avg. number of shots	2.4	2.1	2.2	1.5	1.0	1.9

Table 2: DREAM-1K data statistics.

To quantitatively assess the true capabilities of models, we introduce DREAM-1K (**D**escription with **R**ich **E**vents, **A**ctions, and **M**otions), a collection of 1,000 annotated video clips with diverse complexities. We sourced these videos from five different origins: live-action movies, animated movies, stock videos, long YouTube videos, and TikTok-style short videos, with each category contributing 200 clips. We manually pick all videos from Internet to ensure that the entire dataset is disjoint from public video-language training data. Each clip in DREAM-1K includes at least one dynamic event that cannot be accurately identified from a single frame alone. Table 2 shows the statistics for the dataset. On average, each video lasts 8.9 seconds, includes 6.3 events, 2.2 subjects (people, animals, etc.) and 1.9 shots. We provide a fine-grained manual annotation for each video, covering all events, actions, and motions, with an average length of 59.3 words.

Figure 4 illustrates an example video from DREAM-1K. The video is 14-second long and features six actions. From this example, a noticeable performance gap emerges between models. The descriptions generated by existing models are less comprehensive, often missing key events, or less accurate, introducing hallucinations. In contrast, the descriptions generated by our Tarsier model capture more events and exhibit far fewer hallucinations.



Reference	On the wooden floor, a person sits cross-legged in front of a square cushion, first picking up headphones and putting them on , then picking up the drink beside the square cushion and taking a sip through the straw before placing it back in its original spot . Lastly, the scene concludes with this person bringing both hands together in a gesture of salute or prayer .
VideoChat2	In the video, we see a woman sitting on the floor with her legs crossed , wearing a gray shirt and green pants. She is holding a water bottle and a pair of headphones in her hands . The woman then puts on the headphones and starts to exercise . She is using a pair of pink dumbbells to work out her arms and legs . The video is shot from an aerial view, giving us a clear view of the woman's movements. Overall, the video shows a woman engaging in a workout routine while listening to music.
PLLaVA-34B	In the video, a person is sitting on the floor with a cell phone in front of them. They are holding a pink object , which appears to be a pair of dumbbells. The person is also holding a bottle , which could be a water bottle or a sports drink. The background of the video is a wooden floor, and there is a yoga mat on the floor as well. The person seems to be engaged in some form of exercise or workout, as they are holding the dumbbells and the bottle, which are commonly used during such activities.
GPT-4V	The video clip shows a person sitting on a yoga mat on a wooden floor with fitness equipment: a pair of pink dumbbells, a mobile phone, a pair of headphones, and a bottle of orange liquid. The person is wearing dark pants and a black hoodie. Initially, the person is tying their hair . Then they put on the headphones , pick up the bottle , unscrew the cap , and take a drink . The camera maintains a top-down view throughout the sequence.
Gemini 1.5 Pro	A woman is preparing for workout. She is wearing green pants and a black long sleeve. She is sitting on a gray yoga mat , which has a cellphone and two pink weights on it. A jar with an orange liquid inside and a straw are placed beside the mat. The camera is positioned from the top. The woman reaches up and puts on her headphones . She moves her hands in and out of the shot. Next, she grabs the jar with the orange liquid and takes a drink. After that, she places the jar down and rubs her hands together . Lastly, she places her hands in her lap, palms together with her fingers pointed upwards .
Tarsier-34B (Ours)	A person is sitting on a yoga mat on a wooden floor. There is a smartphone, a pair of pink dumbbells, and a glass of orange juice next to them. The person puts on headphones , then picks up the glass of orange juice , takes a sip, and puts it back down . Finally, the person clasps their hands together .

Figure 3: This video features six actions, each highlighted in a unique color. The Tarsier descriptions capture more actions and contains fewer hallucinations (indicated by underlining and red color) compared to existing models.

4.2 AutoDQ: Automatic evaluation of description quality

Traditional n -gram based metrics, such as CIDEr [67], fall short when evaluating long video descriptions. This is primarily because there are numerous ways to produce a high-quality description, many of which have minimal n -gram overlap with the reference description. Additionally, manually assigning quality scores to descriptions is both costly and subjective. To address this, Maaz et al. [44] propose using ChatGPT to rate descriptions on a scale from 1 to 5. However, the meaning of each rating is ambiguous, and the ratings themselves are not calibrated.

We propose AutoDQ (**A**utomatic **D**escription **Q**uality) as a more interpretable approach to automatic evaluation. Given a reference description D_{ref} and a model-generated description D_{model} , our method compares them through a two-step procedure. In the first step, we use an *event extraction model* to extract a sequence of events from both descriptions. In the second step, we employ a *natural language inference model* to compute two ratios: the ratio of events in D_{ref} that are entailed by D_{model} , and the ratio of events in D_{model} that are entailed by D_{ref} . The first ratio is defined as the recall of D_{model} , while the second ratio is defined as its precision. See Figure 4 for a concrete example. Since both event extraction and natural language inference are well-studied NLP tasks, we can rely on ChatGPT to perform them effectively. For the specific prompts used in

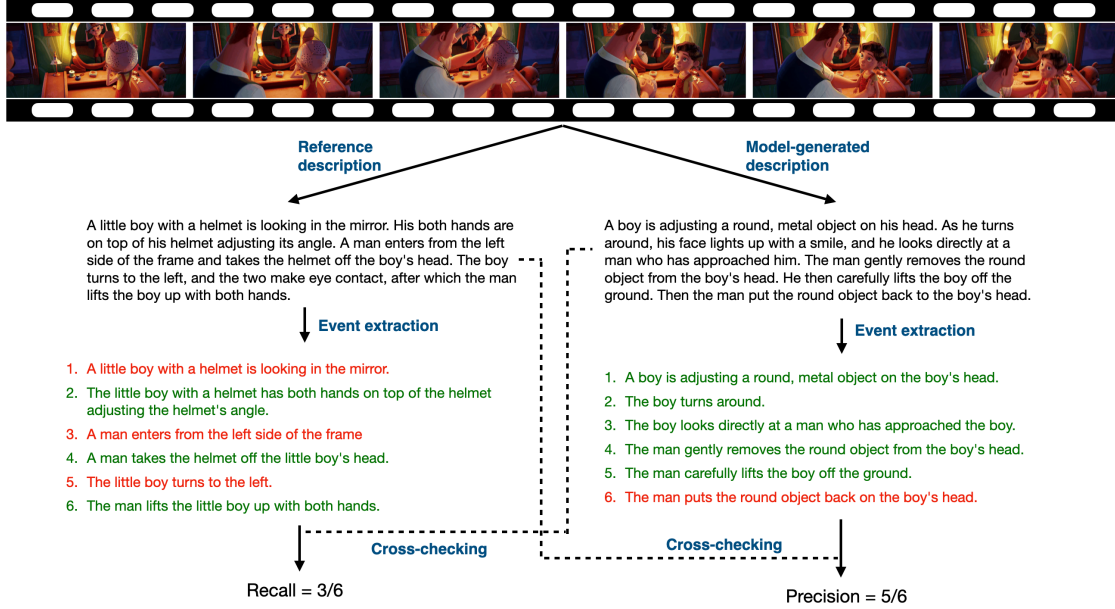


Figure 4: The AutoDQ workflow. AutoDQ uses an extraction model to extract events from two video descriptions, then uses an entailment model to examine how many events extracted from one description are entailed by the other description. We use ChatGPT to implement both models. (Source movie: *Fireheart*, 2022)

these tasks, please refer to Appendix B. Finally, by aggregating these results over all examples, we obtain the precision and recall of the model across the entire dataset, from which we can compute its F1 score.

5 Experiments

In this section, we compare the Tarsier models with state-of-the-art open-source and proprietary models on a variety of video understanding tasks.

5.1 Results on DREAM-1K

In this test, we generate detailed descriptions for each video in DREAM-1K. By default, for Tarsier and GPT-4V, we uniformly sample 8 frames per video. For Gemini 1.5 Pro, it dynamically samples frames at 1 FPS. For other open-source models, we sample frames following their official settings: 8 frames for Video-LLaVA; 90 frames for MiniGPT-4V; 32 frames for LLaVA-NeXT-Video; 16 frames for Tarsier2-7B, GPT-4o [52], VideoChat2 and PLLaVA.

The default prompt is “Describe the video in detail.” For GPT-4V, GPT-4o and Gemini, we found that more specific instructions yield better descriptions, leading us to use a different prompt for these models. For other models, we use the default prompt unless the official implementation recommends a specific one [82, 91]. See Appendix C for the specific prompts that we have used.

Model	Live-action	Animation	YouTube	Shorts	Stock	Overall
Video-LLaVA [35]	19.4/24.3/16.2	15.3/21.2/11.9	21.2/31.9/15.8	18.5/29.4/13.5	27.0/33.5/22.7	20.4/28.1/16.0
MiniGPT-4V [†] [4]	23.6/23.8/23.4	16.4/19.0/14.5	24.8/28.4/22.1	23.7/29.1/20.0	30.7/30.3/31.1	24.0/26.1/22.2
LLaVA-NeXT-Video [*] [91]	25.8/33.4/21.0	22.0/32.0/16.7	26.5/36.2/20.9	24.2/39.8/17.4	31.1/37.3/26.7	26.1/35.7/20.5
VideoChat2 [‡] [32]	26.0/28.1/24.3	18.8/23.9/15.5	27.3/32.7/23.4	26.5/36.8/20.7	33.2/33.6/32.7	26.6/31.0/23.3
PLLaVA-34B [82]	29.3/34.9/25.2	20.9/32.0/15.6	28.9/40.8/22.3	25.6/41.9/18.4	35.1/42.5/29.9	28.2/38.4/22.3
GPT-4V [51]	34.8/39.2/31.3	27.4/31.9/24.0	33.8/40.1/29.2	34.8/46.1/28.0	40.7/46.7/36.1	34.4/40.8/29.7
Gemini 1.5 Pro [58]	36.4/36.4/36.4	30.7/31.8/29.7	34.0/36.7/31.6	37.0/42.4/32.7	42.2/40.7/43.8	36.2/37.6/34.8
GPT-4o [52]	39.8/42.1/37.8	35.8/ 39.1 /33.1	33.8/40.1/29.2	39.9/ 47.9 /34.2	44.0/46.6 /41.7	39.2/ 43.4 /35.7
Tarsier-7B	36.6/38.5/34.8	29.3/34.6/25.5	33.0/39.2/28.4	33.6/44.6/26.9	39.6/44.7/35.5	34.6/40.3/30.2
Tarsier-34B	38.5/39.6/37.5	32.2/35.8/29.2	34.5/41.1 /29.7	34.0/44.1/27.7	41.7/46.4/37.8	36.3/41.4/32.4
Tarsier2-7B	43.8/43.1/44.5	37.1 /37.2/ 37.1	34.5 /37.7/ 31.8	40.9 /45.5/ 37.1	43.9/44.2/ 43.6	40.1 /41.5/ 38.8

Table 3: DREAM-1K evaluation by AutoDQ. We report F1/Precision/Recall scores for each category and for the overall dataset. [†]: MiniGPT-4V-Mistral. [‡]: VideoChat2-Mistral. ^{*}: LLaVA-NeXT-Video-34B-DPO. The Tarsier models achieve significantly higher scores than all open source models in all categories.

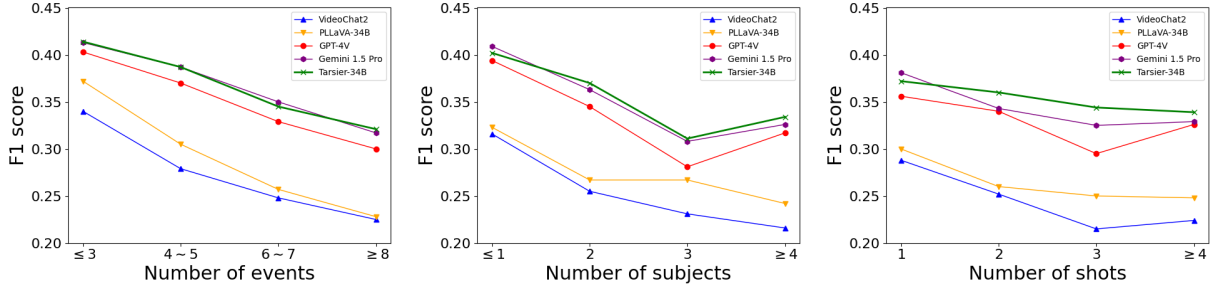


Figure 5: Description quality (F1 score) as a function of increasing video complexity.

5.1.1 Automatic evaluation

First, we report the automatic evaluation results computed by the method in Section 4.2 in Table 3. See the leaderboard for evaluation results of more latest VLMs. The ChatGPT model used to implement AutoDQ is `gpt-3.5-turbo-0125`. As shown in Table 3, the Tarsier models outperform all open-source models in terms of precision, recall, and F1 scores across all categories. Among the existing open-source models, the PLLaVA-34B model exhibits the best overall description quality, yet there is a large gap between its scores and those of Tarsier models. Both the Tarsier-7B and Tarsier-34B models achieve higher F1 scores than GPT-4V. The 34B model also outperforms Gemini 1.5 Pro in the overall score, though the 7B model is slightly weaker than Gemini. This is a promising result, especially considering that our training procedure utilizes significantly less compute resources than these proprietary models. When built upon SigLIP and Qwen2, with a 5x smaller model scale, Tarsier2-7B performs much better than Tarsier-34B by gaining a 40.1 overall F1 score, which even slightly better than the best proprietary model GPT-4o.

Next, we partitioned DREAM-1K according to the number of events, subjects, and shots to test the models against increasing video complexities. As shown in Figure 5, while F1 scores generally decline for all models as video complexity increases, Tarsier-34B consistently outperforms other open-source models and GPT4-V across all levels of complexity, while achieving comparable scores consistently with Gemini 1.5 Pro.

Comparison	Tarsier wins	Same	Baseline wins	Advantage
Tarsier-34B vs PLLaVA-34B	71.7%	8.0%	20.3%	+51.4%
Tarsier-34B vs GPT-4V	50.0%	12.3%	37.7%	+12.3%
Tarsier-34B vs Gemini 1.5 Pro	28.0%	37.3%	34.7%	−6.7%
Tarsier2-7B vs GPT-4o	23.2%	58.4%	18.4%	+4.8%

Table 4: Side-by-side human evaluation results on DREAM-1K. The advantage rate is defined as the win rate subtracted by the loss rate.

5.1.2 Human evaluation

Human side-by-side comparison is a gold standard for evaluating video descriptions. In this test, we randomly sampled 300 videos from DREAM-1K, then asked experienced annotators to compare the descriptions generated by two different models, collecting their preferences. Every pair of descriptions was randomly shuffled to ensure that the annotators were blind to the corresponding models. We compared the Tarsier-34B model against PLLaVA-34B, GPT4-V, and Gemini 1.5 Pro. The results, as shown in Table 4, align with our automatic evaluation findings. Our model significantly outperforms PLLaVA, demonstrating strong capabilities in describing dynamic events with less hallucination. It is also preferred over GPT-4V with a notable margin. When compared to Gemini 1.5 Pro, Tarsier has a slightly negative advantage rate, but in a significant percentage of cases (37.3%), the two models are labeled as “Same”. For Tarsier2-7B, it gains a slight advantage (4.8%) over GPT-4o, and ties with GPT-4o on over one half cases. This indicates that their performance is comparable in many instances. See Appendix D for more information about the human evaluation process.

5.2 Results on other public benchmarks

We evaluated models on popular video understanding benchmarks. As Tarsier is a generalist model, we report its zero-shot performance, meaning that the training sets of these datasets (if they exist) were not used in any stage of training. The results for existing models were sourced directly from their respective papers. Tarsier models sampled 8 frames per video, except for the long-form video understanding benchmark, EgoSchema, where we sampled 16 frames per video.

5.2.1 Multi-choice video question answering

In multi-choice video question answering, the model is given a video, a question, and several answer choices, and is asked to select the correct answer. We compare models on MVBench [32], Next-QA [77], and EgoSchema [45]. MVBench is a comprehensive video understanding benchmark covering 20 tasks; Next-QA is a benchmark for causal and temporal action reasoning; EgoSchema is a benchmark for evaluating long video understanding capabilities. To prevent data leakage, we have removed all EgoSchema-related videos from Ego4D [20] during training.

Results presented in Table 5a demonstrate that the Tarsier models have established new state-of-the-art performance across three benchmarks. We observe a consistent increase in accuracy as the size of the LLM increases, which we attribute to the enhanced reasoning capabilities of larger LLMs.

Model	MVBench	Next-QA	EgoSchema
ST-LLM [40]	54.9	-	-
PLLaVA [82]	58.1	-	-
VideoChat2 [32]	60.4	-	63.6/54.4
InternVideo2 [73]	60.9	-	-
IG-VLM [26]	-	70.9	-
LLoVi [89]	-	67.7	57.6/50.3
VideoAgent [71]	-	71.3	60.2/54.1
LangRepo [25]	-	60.9	66.2/41.2
VideoTree [76]	-	73.5	66.2/61.1
Tarsier-7B	62.6	71.6	56.0/49.9
Tarsier-34B	67.6	79.2	68.6/61.7

(a) Accuracy of zero-shot multi-choice video question answering. On EgoSchema, we report the results on subset/fullset respectively.

Model	MSVD	MSR-VTT	VATEX
BLIP2 [30]	27.4	19.0	-
Panda [14]	49.2	31.5	-
VideoCoca [84]	-	27.1	22.8
Video-LLaMA [90]	47.0	29.1	-
Flamingo [2]	-	-	39.5
VideoPrism [92]	-	38.5	31.7
InternVideo2 [73]	93.1	43.5	49.2
Tarsier-7B	98.0	24.1	51.1
Tarsier-34B	125.9	44.7	57.1

(b) The CIDEr scores ($\uparrow$) of zero-shot video captioning.

Model	MSVD-QA	MSR-VTT-QA	ActivityNet-QA	TGIF-QA
Video-LLaVA [35]	70.7/3.9	59.2/3.5	45.3/3.3	70.0/4.0
MiniGPT-4V [4]	73.9/4.1	58.3/3.5	44.3/3.4	72.2/4.1
VideoChat2 [32]	70.0/3.9	54.1/3.3	49.1/3.3	-
ST-LLM [40]	74.6/3.9	63.2/3.4	50.9/3.3	-
IG-VLM LLaVA-34B [26]	79.6/4.1	62.4/3.5	58.4/3.5	79.1/4.2
IG-VLM GPT-4V [26]	76.3/4.0	63.8/3.5	57.0/3.5	65.3/3.7
Elysium [68]	75.8/3.7	67.5/3.2	43.4/2.9	66.6/3.6
PLLaVA-7B [82]	76.6/4.1	62.0/3.5	56.3/3.5	77.5/4.1
PLLaVA-34B [82]	79.9/4.2	68.7/3.8	60.9/3.7	80.6/4.3
Tarsier-7B	77.0/4.1	62.0/3.5	59.5/3.6	79.2/4.2
Tarsier-34B	80.3/4.2	66.4/3.7	61.6/3.7	82.5/4.4

(c) Accuracy and quality score of zero-shot open-ended video question answering. We employ ChatGPT to evaluate the scores.

Table 5: Zero-shot results on public benchmarks. When citing a baseline model that has multiple variants (e.g., different sizes), we always report the results of the best-performing variant.

5.2.2 Open-ended video question answering

In open-choice video question answering, the model is provided with a video and a question and is asked to generate a free-form answer. We report the results on four standard benchmarks: MSVD-QA [78], MSR-VTT-QA [78], ActivityNet-QA [87], and TGIF-QA [24]. Following the approach of Video-ChatGPT [44], we utilize ChatGPT to assess performance, using the same model version (gpt-3.5-turbo-0125) and the same evaluation prompt as previous work to ensure comparability of results.

The results presented in Table 5c show that Tarsier-34B outperforms all existing models on MSVD-QA, ActivityNet-QA, and TGIF-QA, establishing new state-of-the-art results. Additionally, Tarsier-7B outperforms all existing models on these datasets, with the exception of PLLaVA-34B. Similar to multi-choice VQA, we observe a consistent increase in accuracy and quality as the size of the LLM increases.

5.2.3 Zero-shot video captioning

In this experiment, we evaluated our model on MSVD [9], MSR-VTT [81], and VATEX [72] video captioning benchmarks. Since our model was not exposed to the training sets of these benchmarks, we compared our results with previous zero-shot video captioning models. Table 5b demonstrates that our Tarsier models established new state-of-the-art for zero-shot video captioning. It is important to note that CIDEr scores for zero-shot evaluation are typically much lower than those for models specifically fine-tuned on these datasets. However, fine-tuned models are optimized to match specific benchmarks in terms of caption length, word usage, and language style, which can boost the CIDEr scores but does not necessarily reflect the true description quality.

5.3 Ablation studies

Model	Original setting	Ablating setting	DREAM-1K	MVBench
Tarsier-7B	-	-	34.6	62.6
w/o pre-training	stage1+stage2	stage2 only	30.6	46.6
w/o instruction tuning	stage1+stage2	stage1 only	25.8	60.2
w/o multi-task pre-train	stage1: multi-task data	stage1: WebVid-10M	31.4	48.4
w/o multi-grained description data	stage2: multi-grained description data	stage2: VideoChat2 [32] training data	22.4	61.0
w/o full LLM tuning	train all LLM layers	train top-8 LLM layers	29.9	49.3

Table 6: Ablation study results. We report the F1 scores on DREAM-1K and the accuracy scores on MVBench.

In this section, we conduct ablation studies to assess the effectiveness of our training methodology. We examine the impact of five different ablation settings: (1) omitting pre-training, (2) omitting instruction tuning, (3) substituting our 13.6M multi-task pre-training data with the WebVid-10M [6] video captioning data, (4) substituting our 500K instruction tuning data with the VideoChat2 [32] training data, which contains 788K video instruction tuning pairs with only 7K detailed video descriptions, and (5) rather than tuning all parameters of the LLM, we train only its top-8 layers to reduce computational costs. Due to the high computational costs involved, our experiments are conducted on the 7B model.

Table 6 presents the results of our ablation studies, revealing that each modification leads to a significant decline in performance across both DREAM-1K and MVBench. Specifically, removing stage 1 pre-training or replacing the pre-training dataset with WebVid-10M results in a substantial drop in MVBench accuracy, with decreases of -16.0 and -14.2 absolute points respectively. The table also shows a notable drop in the DREAM-1K scores. These results underscore the importance of multi-task pre-training in developing a robust generalist model. On the other hand, omitting instruction tuning or changing the instruction tuning data significantly diminishes the DREAM-1K score by -8.8 and -12.2 absolute points, highlighting the critical role of our human-annotated, multi-grained video description data in enhancing the model’s video description capabilities. Furthermore, training only the top-8 layers of the LLM leads to inferior outcomes on both datasets, emphasizing the importance of full model tuning.

6 Conclusion and Future Work

In this paper, we release the Tarsier family of models, which outperforms existing open-source video description models in both automatic and human evaluations. Our ablation studies reveal that the strong performance of Tarsier can be attributed to its extensive multi-task pre-training, as well as its fine-tuning on human-annotated, multi-grained video description data. Looking ahead, several directions for further exploration appear promising. First, there is potential for scaling up the pre-training data to the next level, which would require innovative methods to acquire video-text pairs from the internet. Second, scaling up both the visual encoder and the LLM simultaneously could yield improvements. Third, refining the model’s ability to follow complex instructions could further enhance its utility.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [2] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35:23716–23736, 2022.
- [3] Lisa Anne Hendricks, Oliver Wang, Eli Shechtman, Josef Sivic, Trevor Darrell, and Bryan Russell. Localizing moments in video with natural language. In *Proceedings of the IEEE international conference on computer vision*, pages 5803–5812, 2017.
- [4] Kirolos Ataallah, Xiaoqian Shen, Eslam Abdelrahman, Essam Sleiman, Deyao Zhu, Jian Ding, and Mohamed Elhoseiny. Minigpt4-video: Advancing multimodal llms for video understanding with interleaved visual-textual tokens. *arXiv preprint arXiv:2404.03413*, 2024.
- [5] George Awad, Keith Curtis, Asad Butt, Jonathan Fiscus, Afzal Godil, Yooyoung Lee, Andrew Delgado, Eliot Godard, Lukas Diduch, Jeffrey Liu, et al. An overview on the evaluated video retrieval tasks at trecvid 2022. *arXiv preprint arXiv:2306.13118*, 2023.
- [6] Max Bain, Arsha Nagrani, Gül Varol, and Andrew Zisserman. Frozen in time: A joint video and image encoder for end-to-end retrieval. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1728–1738, 2021.
- [7] Satanjeev Banerjee and Alon Lavie. Meteor: An automatic metric for mt evaluation with improved correlation with human judgments. In *Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization*, pages 65–72, 2005.
- [8] Joao Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the kinetics dataset. In *proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6299–6308, 2017.

- [9] David Chen and William B Dolan. Collecting highly parallel data for paraphrase evaluation. In *Proceedings of the 49th annual meeting of the association for computational linguistics: human language technologies*, pages 190–200, 2011.
- [10] Lin Chen, Jisong Li, Xiaoyi Dong, Pan Zhang, Conghui He, Jiaqi Wang, Feng Zhao, and Dahua Lin. Sharegpt4v: Improving large multi-modal models with better captions. *arXiv preprint arXiv:2311.12793*, 2023.
- [11] Sihan Chen, Xingjian He, Longteng Guo, Xinxin Zhu, Weining Wang, Jinhui Tang, and Jing Liu. Valor: Vision-audio-language omni-perception pretraining model and dataset. *arXiv preprint arXiv:2304.08345*, 2023.
- [12] Sihan Chen, Xingjian He, Handong Li, Xiaojie Jin, Jiashi Feng, and Jing Liu. Cosa: Concatenated sample pretrained vision-language foundation model. *arXiv preprint arXiv:2306.09085*, 2023.
- [13] Sihan Chen, Handong Li, Qunbo Wang, Zijia Zhao, Mingzhen Sun, Xinxin Zhu, and Jing Liu. Vast: A vision-audio-subtitle-text omni-modality foundation model and dataset. *Advances in Neural Information Processing Systems*, 36, 2024.
- [14] Tsai-Shien Chen, Aliaksandr Siarohin, Willi Menapace, Ekaterina Deyneka, Hsiang-wei Chao, Byung Eun Jeon, Yuwei Fang, Hsin-Ying Lee, Jian Ren, Ming-Hsuan Yang, et al. Panda-70m: Captioning 70m videos with multiple cross-modality teachers. *arXiv preprint arXiv:2402.19479*, 2024.
- [15] Ho Kei Cheng, Seoung Wug Oh, Brian Price, Alexander Schwing, and Joon-Young Lee. Tracking anything with decoupled video segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1316–1326, 2023.
- [16] Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E. Gonzalez, Ion Stoica, and Eric P. Xing. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality, March 2023.
- [17] Dave Epstein, Boyuan Chen, and Carl Vondrick. Oops! predicting unintentional action in video. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 919–929, 2020.
- [18] Chenyou Fan. Egovqa-an egocentric video question answering benchmark dataset. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, pages 0–0, 2019.
- [19] Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. The” something something” video database for learning and evaluating visual common sense. In *Proceedings of the IEEE international conference on computer vision*, pages 5842–5850, 2017.
- [20] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, Xingyu Liu, et al. Ego4d: Around the

- world in 3,000 hours of egocentric video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18995–19012, 2022.
- [21] Chunhui Gu, Chen Sun, David A Ross, Carl Vondrick, Caroline Pantofaru, Yeqing Li, Sudheendra Vijayanarasimhan, George Toderici, Susanna Ricco, Rahul Sukthankar, et al. Ava: A video dataset of spatio-temporally localized atomic visual actions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6047–6056, 2018.
 - [22] Xingjian He, Sihan Chen, Fan Ma, Zhicheng Huang, Xiaojie Jin, Zikang Liu, Dongmei Fu, Yi Yang, Jing Liu, and Jiashi Feng. Vlab: Enhancing video language pre-training by feature adapting and blending. *arXiv preprint arXiv:2305.13167*, 2023.
 - [23] Ting-Hao Huang, Francis Ferraro, Nasrin Mostafazadeh, Ishan Misra, Aishwarya Agrawal, Jacob Devlin, Ross Girshick, Xiaodong He, Pushmeet Kohli, Dhruv Batra, et al. Visual storytelling. In *Proceedings of the 2016 conference of the North American chapter of the association for computational linguistics: Human language technologies*, pages 1233–1239, 2016.
 - [24] Yunseok Jang, Yale Song, Youngjae Yu, Youngjin Kim, and Gunhee Kim. Tgif-qa: Toward spatio-temporal reasoning in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2758–2766, 2017.
 - [25] Kumara Kahatapitiya, Kanchana Ranasinghe, Jongwoo Park, and Michael S Ryoo. Language repository for long video understanding. *arXiv preprint arXiv:2403.14622*, 2024.
 - [26] Wonkyun Kim, Changin Choi, Wonseok Lee, and Wonjong Rhee. An image grid can be worth a video: Zero-shot video question answering using a vlm. *arXiv preprint arXiv:2403.18406*, 2024.
 - [27] Ranjay Krishna, Kenji Hata, Frederic Ren, Li Fei-Fei, and Juan Carlos Niebles. Dense-captioning events in videos. In *International Conference on Computer Vision (ICCV)*, 2017.
 - [28] Hildegard Kuehne, Hueihan Jhuang, Estíbaliz Garrote, Tomaso Poggio, and Thomas Serre. Hmdb: a large video database for human motion recognition. In *2011 International conference on computer vision*, pages 2556–2563. IEEE, 2011.
 - [29] Jie Lei, Tamara L Berg, and Mohit Bansal. Revealing single frame bias for video-and-language learning. *arXiv preprint arXiv:2206.03428*, 2022.
 - [30] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pages 19730–19742. PMLR, 2023.
 - [31] KunChang Li, Yinan He, Yi Wang, Yizhuo Li, Wenhai Wang, Ping Luo, Yali Wang, Limin Wang, and Yu Qiao. Videochat: Chat-centric video understanding. *arXiv preprint arXiv:2305.06355*, 2023.
 - [32] Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. Mvbench: A comprehensive multi-modal video understanding benchmark. *arXiv preprint arXiv:2311.17005*, 2023.

- [33] Yuncheng Li, Yale Song, Liangliang Cao, Joel Tetreault, Larry Goldberg, Alejandro Jaimes, and Jiebo Luo. Tgif: A new dataset and benchmark on animated gif description. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4641–4650, 2016.
- [34] W Lian, B Goodson, E Pentland, et al. Openorca: An open dataset of gpt augmented flan reasoning traces, 2023.
- [35] Bin Lin, Bin Zhu, Yang Ye, Munan Ning, Peng Jin, and Li Yuan. Video-llava: Learning united visual representation by alignment before projection. *arXiv preprint arXiv:2311.10122*, 2023.
- [36] Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81, 2004.
- [37] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13*, pages 740–755. Springer, 2014.
- [38] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. *arXiv preprint arXiv:2310.03744*, 2023.
- [39] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge, January 2024.
- [40] Ruyang Liu, Chen Li, Haoran Tang, Yixiao Ge, Ying Shan, and Ge Li. St-llm: Large language models are effective temporal learners. *arXiv preprint arXiv:2404.00308*, 2024.
- [41] Yi Liu, Limin Wang, Yali Wang, Xiao Ma, and Yu Qiao. Fineaction: A fine-grained video dataset for temporal action localization. *IEEE transactions on image processing*, 31:6937–6950, 2022.
- [42] Ze Liu, Jia Ning, Yue Cao, Yixuan Wei, Zheng Zhang, Stephen Lin, and Han Hu. Video swin transformer. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3202–3211, 2022.
- [43] Ruipu Luo, Ziwang Zhao, Min Yang, Junwei Dong, Minghui Qiu, Pengcheng Lu, Tao Wang, and Zhongyu Wei. Valley: Video assistant with large language model enhanced ability. *arXiv preprint arXiv:2306.07207*, 2023.
- [44] Muhammad Maaz, Hanoona Rasheed, Salman Khan, and Fahad Shahbaz Khan. Video-chatgpt: Towards detailed video understanding via large vision and language models. *arXiv preprint arXiv:2306.05424*, 2023.
- [45] Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. *Advances in Neural Information Processing Systems*, 36, 2024.
- [46] Joanna Materzynska, Guillaume Berger, Ingo Bax, and Roland Memisevic. The jester dataset: A large-scale video dataset of human gestures. In *Proceedings of the IEEE/CVF international conference on computer vision workshops*, pages 0–0, 2019.

- [47] Antoine Miech, Jean-Baptiste Alayrac, Ivan Laptev, Josef Sivic, and Andrew Zisserman. Rareact: A video dataset of unusual interactions. *arXiv preprint arXiv:2008.01018*, 2020.
- [48] Antoine Miech, Dimitri Zhukov, Jean-Baptiste Alayrac, Makarand Tapaswi, Ivan Laptev, and Josef Sivic. Howto100m: Learning a text-video embedding by watching hundred million narrated video clips. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 2630–2640, 2019.
- [49] Mathew Monfort, SouYoung Jin, Alexander Liu, David Harwath, Rogerio Feris, James Glass, and Aude Oliva. Spoken moments: Learning joint audio-visual representations from video descriptions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14871–14881, 2021.
- [50] Mathew Monfort, Bowen Pan, Kandan Ramakrishnan, Alex Andonian, Barry A McNamara, Alex Lascelles, Quanfu Fan, Dan Gutfreund, Rog rio Schmidt Feris, and Aude Oliva. Multimoments in time: Learning and interpreting models for multi-action video understanding. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(12):9434–9445, 2021.
- [51] OpenAI. Gpt-4v system card. <https://openai.com/index/gpt-4v-system-card/>, 2023.
- [52] OpenAI. Gpt-4o system card. <https://openai.com/index/hello-gpt-4o/>, 2024.
- [53] AJ Piergiovanni, Weicheng Kuo, and Anelia Angelova. Rethinking video vits: Sparse video tubes for joint image and video learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2214–2224, 2023.
- [54] AJ Piergiovanni, Isaac Nobel, Dahun Kim, Michael S Ryoo, Victor Gomes, and Anelia Angelova. Mirasol3b: A multimodal autoregressive model for time-aligned and contextual modalities. *arXiv preprint arXiv:2311.05698*, 2023.
- [55] Bryan A Plummer, Liwei Wang, Chris M Cervantes, Juan C Caicedo, Julia Hockenmaier, and Svetlana Lazebnik. Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models. In *Proceedings of the IEEE international conference on computer vision*, pages 2641–2649, 2015.
- [56] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021.
- [57] Michaela Regneri, Marcus Rohrbach, Dominikus Wetzels, Stefan Thater, Bernt Schiele, and Manfred Pinkal. Grounding action descriptions in videos. *Transactions of the Association for Computational Linguistics*, 1:25–36, 2013.
- [58] Machel Reid, Nikolay Savinov, Denis Teplyashin, Dmitry Lepikhin, Timothy Lillicrap, Jean-Baptiste Alayrac, Radu Soricut, Angeliki Lazaridou, Orhan Firat, Julian Schrittwieser, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.

- [59] Anna Rohrbach, Atousa Torabi, Marcus Rohrbach, Niket Tandon, Christopher Pal, Hugo Larochelle, Aaron Courville, and Bernt Schiele. Movie description. *International Journal of Computer Vision*, 123:94–120, 2017.
- [60] Gunnar A Sigurdsson, Abhinav Gupta, Cordelia Schmid, Ali Farhadi, and Karteek Alahari. Charades-ego: A large-scale dataset of paired third and first person videos. *arXiv preprint arXiv:1804.09626*, 2018.
- [61] Gunnar A Sigurdsson, Gül Varol, Xiaolong Wang, Ali Farhadi, Ivan Laptev, and Abhinav Gupta. Hollywood in homes: Crowdsourcing data collection for activity understanding. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*, pages 510–526. Springer, 2016.
- [62] Quan Sun, Qiyang Yu, Yufeng Cui, Fan Zhang, Xiaosong Zhang, Yuezhe Wang, Hongcheng Gao, Jingjing Liu, Tiejun Huang, and Xinlong Wang. Generative pretraining in multimodality. *arXiv preprint arXiv:2307.05222*, 2023.
- [63] Yansong Tang, Dajun Ding, Yongming Rao, Yu Zheng, Danyang Zhang, Lili Zhao, Jiwen Lu, and Jie Zhou. Coin: A large-scale dataset for comprehensive instructional video analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1207–1216, 2019.
- [64] Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- [65] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. *Advances in neural information processing systems*, 35:10078–10093, 2022.
- [66] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- [67] Ramakrishna Vedantam, C Lawrence Zitnick, and Devi Parikh. Cider: Consensus-based image description evaluation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4566–4575, 2015.
- [68] Han Wang, Yanjie Wang, Yongjie Ye, Yuxiang Nie, and Can Huang. Elysium: Exploring object-level perception in videos via mllm. *arXiv preprint arXiv:2403.16558*, 2024.
- [69] Jianfeng Wang, Zhengyuan Yang, Xiaowei Hu, Linjie Li, Kevin Lin, Zhe Gan, Zicheng Liu, Ce Liu, and Lijuan Wang. Git: A generative image-to-text transformer for vision and language. *arXiv preprint arXiv:2205.14100*, 2022.
- [70] Jinpeng Wang, Yixiao Ge, Rui Yan, Yuying Ge, Kevin Qinghong Lin, Satoshi Tsutsui, Xudong Lin, Guanyu Cai, Jianping Wu, Ying Shan, et al. All in one: Exploring unified video-language pre-training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6598–6608, 2023.

- [71] Xiaohan Wang, Yuhui Zhang, Orr Zohar, and Serena Yeung-Levy. Videoagent: Long-form video understanding with large language model as agent. *arXiv preprint arXiv:2403.10517*, 2024.
- [72] Xin Wang, Jiawei Wu, Junkun Chen, Lei Li, Yuan-Fang Wang, and William Yang Wang. Vatex: A large-scale, high-quality multilingual dataset for video-and-language research. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4581–4591, 2019.
- [73] Yi Wang, Kunchang Li, Xinhao Li, Jiashuo Yu, Yinan He, Guo Chen, Baoqi Pei, Rongkun Zheng, Jilan Xu, Zun Wang, et al. Internvideo2: Scaling video foundation models for multi-modal video understanding. *arXiv preprint arXiv:2403.15377*, 2024.
- [74] Yi Wang, Kunchang Li, Yizhuo Li, Yinan He, Bingkun Huang, Zhiyu Zhao, Hongjie Zhang, Jilan Xu, Yi Liu, Zun Wang, et al. Internvideo: General video foundation models via generative and discriminative learning. *arXiv preprint arXiv:2212.03191*, 2022.
- [75] Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language models with self-generated instructions. *arXiv preprint arXiv:2212.10560*, 2022.
- [76] Ziyang Wang, Shoubin Yu, Elias Stengel-Eskin, Jaehong Yoon, Feng Cheng, Gedas Bertasius, and Mohit Bansal. Videotree: Adaptive tree-based video representation for llm reasoning on long videos. *arXiv preprint arXiv:2405.19209*, 2024.
- [77] Junbin Xiao, Xindi Shang, Angela Yao, and Tat-Seng Chua. Next-qa: Next phase of question-answering to explaining temporal actions. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9777–9786, 2021.
- [78] Dejing Xu, Zhou Zhao, Jun Xiao, Fei Wu, Hanwang Zhang, Xiangnan He, and Yueting Zhuang. Video question answering via gradually refined attention over appearance and motion. In *Proceedings of the 25th ACM international conference on Multimedia*, pages 1645–1653, 2017.
- [79] Haiyang Xu, Qinghao Ye, Ming Yan, Yaya Shi, Jiabo Ye, Yuanhong Xu, Chenliang Li, Bin Bi, Qi Qian, Wei Wang, et al. mplug-2: A modularized multi-modal foundation model across text, image and video. In *International Conference on Machine Learning*, pages 38728–38748. PMLR, 2023.
- [80] Hu Xu, Gargi Ghosh, Po-Yao Huang, Dmytro Okhonko, Armen Aghajanyan, Florian Metze, Luke Zettlemoyer, and Christoph Feichtenhofer. Videoclip: Contrastive pre-training for zero-shot video-text understanding. *arXiv preprint arXiv:2109.14084*, 2021.
- [81] Jun Xu, Tao Mei, Ting Yao, and Yong Rui. Msr-vtt: A large video description dataset for bridging video and language. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5288–5296, 2016.
- [82] Lin Xu, Yilin Zhao, Daquan Zhou, Zhijie Lin, See Kiong Ng, and Jiashi Feng. Pllava: Parameter-free llava extension from images to videos for video dense captioning. *arXiv preprint arXiv:2404.16994*, 2024.

- [83] Hongwei Xue, Tiankai Hang, Yanhong Zeng, Yuchong Sun, Bei Liu, Huan Yang, Jianlong Fu, and Baining Guo. Advancing high-resolution video-language representation with large-scale video transcriptions. In *International Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [84] Shen Yan, Tao Zhu, Zirui Wang, Yuan Cao, Mi Zhang, Soham Ghosh, Yonghui Wu, and Jiahui Yu. Videococa: Video-text modeling with zero-shot transfer from contrastive captioners. *arXiv preprint arXiv:2212.04979*, 2022.
- [85] An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, Guanting Dong, Haoran Wei, Huan Lin, Jialong Tang, Jialin Wang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Ma, Jianxin Yang, Jin Xu, Jingren Zhou, Jinze Bai, Jinzheng He, Junyang Lin, Kai Dang, Keming Lu, Keqin Chen, Kexin Yang, Mei Li, Mingfeng Xue, Na Ni, Pei Zhang, Peng Wang, Ru Peng, Rui Men, Ruize Gao, Runji Lin, Shijie Wang, Shuai Bai, Sinan Tan, Tianhang Zhu, Tianhao Li, Tianyu Liu, Wenbin Ge, Xiaodong Deng, Xiaohuan Zhou, Xingzhang Ren, Xinyu Zhang, Xipin Wei, Xuancheng Ren, Xuejing Liu, Yang Fan, Yang Yao, Yichang Zhang, Yu Wan, Yunfei Chu, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, Zhifang Guo, and Zhihao Fan. Qwen2 technical report, 2024.
- [86] Kexin Yi, Chuang Gan, Yunzhu Li, Pushmeet Kohli, Jiajun Wu, Antonio Torralba, and Joshua B Tenenbaum. Clevrer: Collision events for video representation and reasoning. *arXiv preprint arXiv:1910.01442*, 2019.
- [87] Zhou Yu, Dejing Xu, Jun Yu, Ting Yu, Zhou Zhao, Yueting Zhuang, and Dacheng Tao. Activitynet-qa: A dataset for understanding complex web videos via question answering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 9127–9134, 2019.
- [88] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training, 2023.
- [89] Ce Zhang, Taixi Lu, Md Mohaiminul Islam, Ziyang Wang, Shoubin Yu, Mohit Bansal, and Gedas Bertasius. A simple llm framework for long-range video question-answering. *arXiv preprint arXiv:2312.17235*, 2023.
- [90] Hang Zhang, Xin Li, and Lidong Bing. Video-llama: An instruction-tuned audio-visual language model for video understanding. *arXiv preprint arXiv:2306.02858*, 2023.
- [91] Yuanhan Zhang, Bo Li, haotian Liu, Yong jae Lee, Liangke Gui, Di Fu, Jiashi Feng, Ziwei Liu, and Chunyuan Li. Llava-next: A strong zero-shot video understanding model, April 2024.
- [92] Long Zhao, Nitesh B Gundavarapu, Liangzhe Yuan, Hao Zhou, Shen Yan, Jennifer J Sun, Luke Friedman, Rui Qian, Tobias Weyand, Yue Zhao, et al. Videoprism: A foundational visual encoder for video understanding. *arXiv preprint arXiv:2402.13217*, 2024.
- [93] Yipin Zhou, Zhaowen Wang, Chen Fang, Trung Bui, and Tamara L Berg. Visual to sound: Generating natural sound for videos in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3550–3558, 2018.

A Training hyperparameters

The following table shows the training hyperparameters in Stage 1 and Stage 2.

Configuration	Stage 1 (Pre-training)	Stage 2 (Instruction Tuning)
ViT init.	clip-vit-large-patch14-336	
LLM init.	LLaVA-NeXT	
VL Adapter init.	LLaVA-NeXT	
Optimizer name	AdamW	AdamW
Optimizer β_1	0.9	0.9
Optimizer β_2	0.999	0.999
Optimizer eps	$1e^{-6}$	$1e^{-6}$
Learning rate	$2e^{-5}$	$2e^{-5}$
Learning rate schedule	cosine	cosine
Training steps	100,000	10,000
Warm-up steps	1,000	1,500
Weight decay	0.01	0.01
Gradient clip	1.0	1.0
Dropout rate	0	0
Global batch size	288	96
Frame resolution	336x336	336x336
Tokens per frame	576	576
Frames per video	[8,32]	{8,16}
Numerical precision	bfloat16	bfloat16

Table 7: Training hyperparameters of Tarsier

B AutoDQ prompts

This appendix presents the ChatGPT prompts used by AutoDQ to execute automatic evaluation. Below is the prompt for event extraction. There is a placeholder to put in the video description text from which events will be extracted:

Below is a description of a video clip:

[The video description]

Extract at most 10 key events from the above video description paragraph.

Requirements:

- An event must include an action, motion or movement (NOT STATIC INFORMATION). DON'T repeat same events.
- Every event is represented by a brief sentence with in 10 words, with a subject, a predicate and optionally an object, avoid unnecessary appearance descriptions.
- Every event must be atomic, meaning that it cannot be further split into multiple events.
- Scene cuts and camera motions are NOT events.
- Substitute pronouns by the nouns they refer to.

Please generate the response in the form of a Python dictionary string with keys "events". The value of "events" is a List(str), of which each item is an event. DO NOT PROVIDE ANY OTHER OUTPUT TEXT OR EXPLANATION. Only provide the Python dictionary string. For example, your response should look like this:{"events": [event1, event2, ...]}

Below is the ChatGPT prompt for predicting entailment. There are placeholders to put in the video description text, as well as the events extracted by the previous prompt:

Given a video description and a list of events. For each event, classify the relationship between the video description and the event into three classes: entailment, neutral, contradiction.

- "entailment" means that the video description entails the event.
- "contradiction" means that some detail in the video description contradicts with the event.
- "neutral" means that the relationship is neither "entailment" or "contradiction".

Output a list in Json format:

```
[ {"event": "copy an event here", "relationship": "put class name here",  
  "reason": "give a reason"}, ... ]
```

Video description:

[The video description]

Events:

[The events in Json list-of-string format]

DO NOT PROVIDE ANY OTHER OUTPUT TEXT OR EXPLANATION. Only output the JSON.

Output:

C Video description prompts

This appendix presents the prompts used for the DREAM-1K experiments in Section 5.1. The default prompt is:

```
Describe the video in detail.
```

The prompt for GPT-4V:

```
Given 8 frames uniformly sampled from a video clip, describe the video (not the individual images!) in detail, focusing on the main subjects, their actions and the background scene. DON'T describe feelings or atmosphere.
```

The prompt for Gemini 1.5 Pro:

```
Describe the video in one paragraph, mainly focusing on the dynamic events in the video. Don't describe feelings or atmosphere.
```

The prompt for PLLaVA-34B, as recommended by [82]:

```
You are to assist me in accomplishing a task about the input video. Reply to me with a precise yet detailed response. For how you would succeed in the recaptioning task, read the following Instructions section and Then, make your response with a elaborate paragraph.

# Instructions
1. Avoid providing over detailed information such as color, counts of any objects as you are terrible regarding observing these details
2. Instead, you should carefully go over the provided video and reason about key information about the overall video
3. If you are not sure about something, do not include it in you response.

# Task
Describe the background, characters and the actions in the provided video.
```

The prompt for LLaVA-NeXT-Video, as recommended by [91]:

```
Please provide a detailed description of the video, focusing on the main subjects, their actions, and the background scenes.
```

D Human evaluation details

In the human evaluation, we randomly sampled a subset of DREAM-1K, consisting of 300 videos. Three experienced annotators were involved in the evaluation. For each model pair, each annotator was presented with 100 video description pairs. Each pair contained two descriptions generated by the two models, but their order was randomly swapped to ensure that the annotator could not identify the model that produced each description. The annotators were asked to assign one of

three labels to each pair: “Description A is better,” “Description B is better,” or “Same quality.” Before the annotation process began, we provided them with the following annotation guide:

Imagine you are reading video descriptions to a blind person. Which version of the description better helps the blind person understand the content of the video? You should choose the more helpful version. This judgment may be subjective, but in principle, you can consider both accuracy and comprehensiveness:

- When equally accurate, the more comprehensive description is more helpful.
- When equally comprehensive, the more accurate description is more helpful.
- If both descriptions have flaws in accuracy and comprehensiveness, you need to judge which flaw has a lesser impact on understanding the main content of the video. In this case, you can use your own subjective feelings and common sense to make a judgment, without relying on explicit rules.
- We value the description of dynamic events. Therefore, your evaluation should focus on dynamic events (actions, behaviors, changes, etc.). Detailed descriptions of static aspects (appearance, color, texture, static background) do not earn extra points, unless they help understand key events. The model does not accept audio input, so you can completely ignore the sounds in the video.
- Please make an effort to discern the better description, and try to use the “Same quality” option as little as possible.